

# **AI-DRIVEN MULTIMODAL SENTIMENT ANALYSIS FOR MENTAL HEALTH MONITORING ON SOCIAL MEDIA**

**A Research Proposal Synopsis  
(Based on Detailed Literature Review)**

Submitted in Partial Fulfillment of the Requirements for the Degree of

**Bachelor of Technology  
in  
Computer Science and Engineering**

Session: **2023–2027**

by

KARTIKEY TIWARI (2300271540063)  
AKSHAT SHARMA (2300271540018)  
KARAN SHARMA (2300271540062)  
LALIT SHARMA (2300271540066)

Under the Supervision of

**Dr.AVDHESH GUPTA**



**AJAY KUMAR GARG ENGINEERING COLLEGE, GHAZIABAD  
DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW**

August 20, 2026

# DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

We also declare that the intellectual content of this synopsis is the product of our own work, even though we may have received assistance from others on style, presentation, and language expression.

We further declare that the research proposal outlined herein is based on an original review of existing literature conducted by us, and that any resulting publication (conference paper, journal article) or intellectual property (patent) arising from this work will duly acknowledge all sources and collaborators.

In accordance with the project rubric's emphasis on **Academic Integrity**, we commit to using plagiarism detection tools like Turnitin for self-checking throughout the writing process to ensure the originality of our work.

| Name of Student | Roll Number | Signature |
|-----------------|-------------|-----------|
|-----------------|-------------|-----------|

|                 |                 |  |
|-----------------|-----------------|--|
| KARTIKEY TIWARI | (2300271540063) |  |
|-----------------|-----------------|--|

|               |                 |  |
|---------------|-----------------|--|
| AKSHAT SHARMA | (2300271540018) |  |
|---------------|-----------------|--|

|              |                 |  |
|--------------|-----------------|--|
| KARAN SHARMA | (2300271540062) |  |
|--------------|-----------------|--|

|              |                 |  |
|--------------|-----------------|--|
| LALIT SHARMA | (2300271540066) |  |
|--------------|-----------------|--|

Session: 2023–2027

Date: August 20, 2026

# CERTIFICATE

This is to certify that the work which is being presented in the B.Tech Research Proposal Synopsis entitled “**AI-Driven Multimodal Sentiment Analysis For Mental Health Monitoring On Social Media**”, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Session 2023–2027) and submitted to Ajay Kumar Garg Engineering College, Ghaziabad, affiliated with Dr. A.P.J. Abdul Kalam Technical University, Lucknow, is an authentic record of our own work carried out under my supervision.

The matter presented in this synopsis, including the literature review, proposed methodology, and the planned research outcome (conference/journal publication or patent), has not been submitted by us for the award of any other degree of this or any other University.

**Dr.AVDHESH GUPTA**

Professor Incharge

Department of Computer Science and Engineering

Ajay Kumar Garg Engineering College, Ghaziabad

# ABSTRACT

The rapid growth of social media has created a valuable source of information for understanding people's emotions and behavioral patterns nowadays. This project proposes an AI-driven multimodal system for analysing social media content to identify emotional states that may indicate potential mental health concerns which would help the experts in preventing the bad happenings. The proposed approach combines Natural Language Processing and Deep Learning to examine different forms of user-generated content, including written posts, captions, emojis, and images. Relevant features are extracted from these modalities and integrated to classify emotional patterns such as happiness, stress, anxiety, sadness, anger, and loneliness. Models such as LSTM, CNN, and transformer-based architectures can be employed for sentiment and emotion classification. The system can further support real-time monitoring through visual dashboards, enabling researchers and professionals to observe emerging behavioral trends. The proposed solution demonstrates how Artificial Intelligence can be applied to social media analytics and mental health monitoring, with potential extensions including multilingual analysis, conversational support, and personalized recommendations.

**Keywords:** Artificial Intelligence, Multimodal Sentiment Analysis, Mental Health Monitoring, Social Media Analytics, Natural Language Processing (NLP), Deep Learning, Emotion Recognition, Sentiment Classification, Machine Learning, LSTM, CNN, BERT, Text Analysis, Image Analysis, Behavioral Pattern Analysis

# Contents

|                                                               |     |
|---------------------------------------------------------------|-----|
| <b>DECLARATION</b>                                            | i   |
| <b>CERTIFICATE</b>                                            | ii  |
| <b>ABSTRACT</b>                                               | iii |
| <b>1 Introduction</b>                                         |     |
| 1.1 Background and Motivation . . . . . 1.1.1                 | 1   |
| DomainOverview .....                                          | 1   |
| 1.1.2 MotivatingScenario . . . . .                            | 1   |
| 1.2 ProblemStatement . . . . .                                | 1   |
| 1.3 Research Questions and Objectives . . .                   | 1   |
| 1.3.1 PrimaryResearchQuestion...                              | 2   |
| 1.3.2 SecondaryResearchQuestions.                             | 2   |
| 1.3.3 ResearchObjectives . . . . .                            | 2   |
| 1.4 Scope of the Project. . . . .                             | 2   |
| <b>2 Literature Review</b>                                    |     |
| 2.1 ThematicReviewofCurrentState-of-the-Art . . . . .         | 3   |
| 2.1.1 Theme 1: Text-Based Mental Health Analysis.....         | 3   |
| 2.1.2 Theme 2: Image-Based Emotion Analysis                   | 3   |
| 2.1.3Theme 3: Multimodal Sentiment and Mental Health Analysis | 3   |
| 2.2 Identification of Research Gap                            | 4   |
| 2.3 Formulation of the Research Proposal                      | 4   |

.. .

|          |                                                                  |          |
|----------|------------------------------------------------------------------|----------|
| <b>3</b> | <b>Societal Impact and SDG Alignment</b>                         | <b>5</b> |
| 3.1      | PrimarySDGRelevance.....                                         | 5        |
| 3.1.1    | MappedSDGandTarget.....                                          | 5        |
| 3.1.2    | AlignmentJustification.....                                      | 5        |
| 3.2      | BreadthofSDGLinkages.....                                        | 5        |
| 3.2.1    | Secondary SDG 1: SDG 9 — Industry, Innovation and Infrastructure | 5        |
| 3.2.2    | Secondary SDG 2: SDG 10 — Reduced Inequalities                   | 6        |
| <b>4</b> | <b>Research Methodology</b>                                      | <b>7</b> |
| 4.1      | ResearchApproach.....                                            | 7        |
| 4.1.1    | JustificationofChosenApproach . . . . .                          | 7        |
| 4.2      | ProposedSystem/Algorithm/Framework . . .                         | 7        |
| 4.2.1    | SystemArchitecture/ConceptualModel                               | 7        |
| 4.2.2    | Algorithmic/DesignDetails . . . . .                              | 8        |
| 4.3      | DataCollectionandAnalysis . . . . .                              | 8        |
| 4.3.1    | DataSources/DatasetDescription....                               | 8        |
| 4.3.2    | AnalysisTechniques ..... . . . .                                 | 8        |
| <b>5</b> | <b>Expected Outcomes and Dissemination Plan</b>                  |          |
| 5.1      | ExpectedResearchContributions. . . . .                           | 9        |
| 5.2      | ModeofDissemination . . . . .                                    | 9        |
| 5.3      | OutcomeMappingTable . . . . .                                    | 9        |
| 5.4      | Anticipated Challenges to Dissemination                          | 10       |
| <b>6</b> | <b>Project Plan and Professional Practices</b>                   |          |
| 6.1      | TimelineandMilestones . . . . .                                  | 11       |
| 6.1.1    | WorkBreakdownStructure.....                                      | 11       |
| 6.2      | CollaborationandDocumentation . . . . .                          | 11       |
| 6.2.1    | TeamRolesandResponsibilities.....                                | 12       |
| 6.2.2    | CollaborativeToolsandVersionControl. . . . .                     | 12       |

# Chapter 1

## Introduction

### 1.1 Background and Motivation

#### 1.1.1 Domain Overview

Artificial Intelligence and Machine Learning techniques are increasingly being used to analyse social media content and identify meaningful patterns. Natural Language Processing enables systems to analyse textual information and determine sentiments, emotions, and linguistic characteristics. However, social media communication also includes images, memes, and other visual information that may contain important emotional context. Multimodal AI addresses this limitation by combining information from multiple modalities such as text and images.

#### 1.1.2 Motivating Scenario

A social media post may contain both text and an image that together express a particular emotional state. A text-only system may understand the written message but ignore important information present in the accompanying image. Similarly, an image-only system may fail to understand the explicit meaning conveyed through the user's words. Therefore, analysing textual and visual information together can provide a more complete representation of the emotional content of a post.

### 1.2 Problem Statement

Existing sentiment and mental-health analysis systems often rely primarily on textual information from social media. Such approaches may lose relevant emotional and contextual information contained in images, memes, and other visual content. The project therefore aims to develop an AI-driven multimodal framework that jointly analyses textual and visual information for sentiment-based mental health monitoring. The proposed system will also compare multimodal analysis with individual text-based and image-based approaches.

## **1.3 Research Questions and Objectives**

### **1.3.1 Primary Research Question**

Can an AI-driven multimodal framework combining textual and visual information improve sentiment-based mental health monitoring on social media compared with unimodal approaches?

### **1.3.2 Secondary Research Questions**

How effectively can textual information be used to identify mental-health-related sentiment in social media posts? What additional information can visual content provide for such classification? Does combining textual and visual features improve classification performance compared with individual modalities?

### **1.3.3 Research Objectives**

The project aims to study existing AI-based approaches for social-media sentiment and mental-health analysis and develop suitable preprocessing methods for textual and visual data. It will develop a multimodal framework that combines text and image representations for classification. The performance of the multimodal approach will be compared with text-only and image-only models. Standard metrics such as accuracy, precision, recall, and F1-score will be used for evaluation.

## **1.4 Scope of the Project**

The project focuses on computational analysis of textual and visual information obtained from social media content. It will include data preprocessing, feature extraction, multimodal feature fusion, classification, and performance evaluation using Artificial Intelligence and Deep Learning techniques. The proposed framework will focus on identifying mental-health-related sentiment or patterns rather than providing clinical diagnosis. The system is intended as a research and monitoring support tool and not as a replacement for qualified mental-health professionals.

# Chapter 2

## Literature Review

### 2.1 Thematic Review of Current State-of-the-Art

#### 2.1.1 Theme 1: Text-Based Mental Health Analysis

Text-based mental-health analysis uses Natural Language Processing and Machine Learning techniques to identify emotional and linguistic patterns in social media posts. Traditional approaches use techniques such as word representations and statistical features, while modern approaches use deep learning and transformer-based models such as BERT. These methods can identify useful textual indicators associated with different sentiments and mental-health-related states. However, they cannot directly utilise information contained in visual content.

#### 2.1.2 Theme 2: Image-Based Emotion Analysis

Images shared on social media can contain emotional and contextual information that may not be explicitly expressed through words. Computer Vision and Deep Learning models such as Convolutional Neural Networks can be used to extract meaningful visual features from such content. Visual analysis can therefore provide additional information for understanding the emotional context of social media posts. However, image-only approaches may fail to understand the textual meaning associated with the visual content.

#### 2.1.3 Theme 3: Multimodal Sentiment and Mental Health Analysis

Multimodal analysis combines information from different data types to obtain a richer representation of social media content. Existing approaches combine textual and visual features using techniques such as feature fusion, attention mechanisms, and multimodal learning architectures. Such approaches can capture complementary information from multiple modalities and may improve classification performance. However, challenges remain in effectively combining modalities and handling differences or conflicts between textual and visual information.

## **2.2 Identification of Research Gap**

Existing research demonstrates the usefulness of both text-based and multimodal approaches for analysing social media content related to sentiment and mental health. However, effectively integrating textual and visual information remains challenging because different modalities may contain complementary or conflicting emotional cues. There is also a need to evaluate multimodal approaches against individual modality-based models using consistent performance measures. This project addresses this gap by developing and evaluating a practical AI-driven framework for multimodal sentiment-based mental health monitoring.

## **2.3 Formulation of the Research Proposal**

Based on the identified research gap, the proposed work will combine textual and visual information from social media posts within a single analytical framework. Textual features will be extracted using suitable NLP techniques, while visual features will be obtained using an appropriate computer-vision model. The extracted features will then be fused and supplied to a classification model for generating the final output. The proposed approach will be evaluated against unimodal approaches to determine the usefulness of multimodal information.

# **Chapter 3**

## **Societal Impact and SDG Alignment**

### **3.1 Primary SDG Relevance**

#### **3.1.1 MappedSDGandTarget**

The primary Sustainable Development Goal associated with this project is SDG 3: Good Health and Well-being. The project supports the broader objective of promoting mental well-being by using Artificial Intelligence to analyse mental-health-related signals expressed through social media. The system is intended to provide computational support for monitoring and research related to mental health.

#### **3.1.2 Alignment Justification**

Mental-health concerns can affect individuals' well-being and quality of life, while social media provides a large source of user-generated emotional information. The proposed multimodal system can analyse textual and visual cues to identify patterns that may be relevant for mental-health monitoring. By supporting earlier computational identification of potentially concerning patterns, the project can contribute to research and awareness related to mental well-being. It is intended to complement, rather than replace, professional assessment.

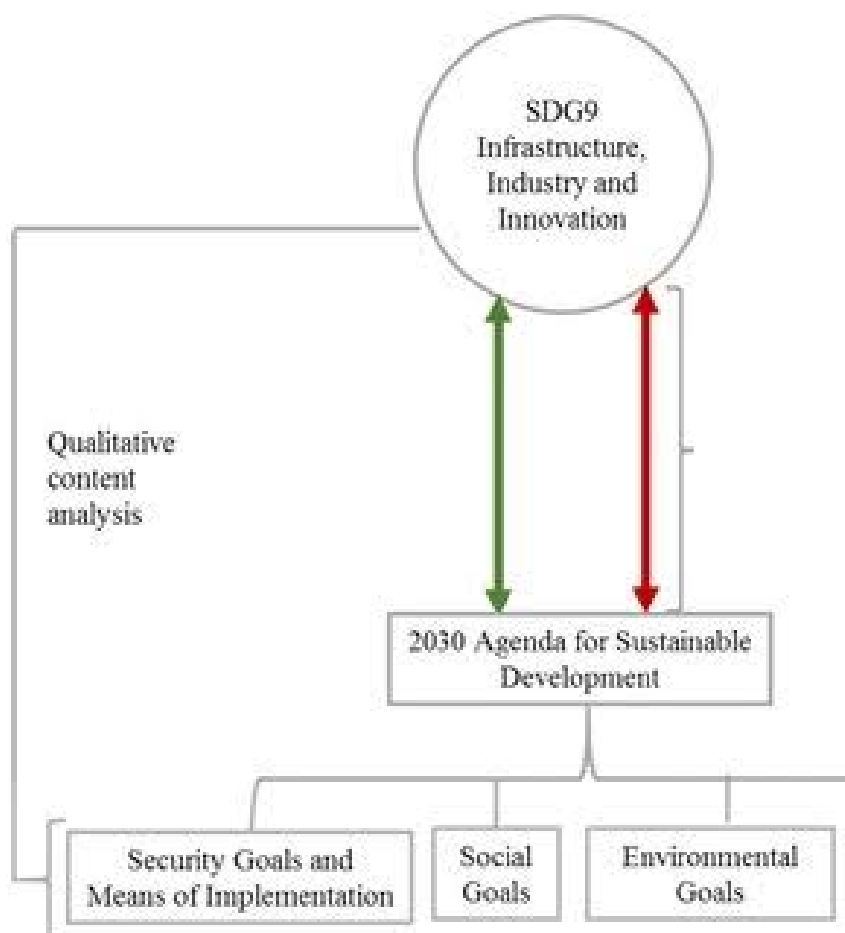
### **3.2 Breadth of SDG Linkages**

#### **3.2.1 Secondary SDG 1: SDG 9 — Industry, Innovation and Infrastructure**

The project supports SDG 9: Industry, Innovation and Infrastructure through the development of an AI-driven multimodal framework. It applies modern Natural Language Processing, Computer Vision, and multimodal learning techniques to a socially relevant problem. The proposed framework demonstrates how emerging AI technologies can be integrated to develop innovative digital solutions.

### 3.2.2 Secondary SDG 2: SDG 10 — Reduced Inequalities

The project can contribute to SDG 10: Reduced Inequalities by exploring computational methods that may support scalable and accessible mental-health monitoring. Digital platforms can provide large-scale sources of information without requiring physical access to a monitoring facility. However, the system should be developed responsibly so that differences in language, culture, digital access, and data representation do not create unfair outcomes.



# **Chapter 4**

## **Research Methodology**

### **4.1 Research Approach**

The project follows an experimental and comparative research approach for developing and evaluating a multimodal AI framework. The study will involve dataset preparation, preprocessing, feature extraction, model development, multimodal fusion, and performance evaluation. Separate unimodal models will also be considered to provide a meaningful comparison with the proposed multimodal approach.

#### **4.1.1 Justification of Chosen Approach**

A comparative experimental approach is suitable because the primary aim is to determine whether combining different modalities provides an improvement over individual modalities. The approach allows textual, visual, and multimodal models to be evaluated using common performance metrics. The resulting comparison will provide quantitative evidence regarding the effectiveness of multimodal analysis.

### **4.2 Proposed System / Algorithm / Framework**

#### **4.2.1 System Architecture / Conceptual Model**

The proposed framework will accept social media content containing textual and visual information as input. The text will undergo Natural Language Processing, while the associated images will be processed using suitable Computer Vision techniques. Features extracted from both modalities will be combined through a feature-fusion mechanism and passed to a classification stage. The final system will generate a mental-health-related sentiment or classification output based on the selected dataset categories.

#### **4.2.2 Algorithmic / Design Details**

The proposed workflow begins with collection and preparation of suitable multimodal social media data. Text will be cleaned and transformed into suitable representations, while images will be resized, normalised, and converted into visual features.

## **4.3 Data Collection and Analysis**

### **4.3.1 Data Sources / Dataset Description**

The dataset will be selected from a publicly available research dataset containing social media posts with corresponding textual and visual information. The selected dataset will be examined for the number of samples, available modalities, class labels, and data quality before model development. Appropriate preprocessing will be performed to remove irrelevant or noisy information. The final dataset selection will depend on the availability of both text and image data and suitable mental-health-related labels.

### **4.3.2 Analysis Techniques**

Model performance will be evaluated using accuracy, precision, recall, and F1-score. A confusion matrix will be used to analyse classification errors across different classes. The performance of the multimodal model will be compared with text-only and image-only approaches. These comparisons will help determine whether combining multiple modalities provides measurable performance improvements.

# **Chapter 5**

## **Expected Outcomes and Dissemination Plan**

### **5.1 Expected Research Contributions**

The project is expected to produce an AI-based framework capable of combining textual and visual information for mental-health-related sentiment analysis. It will provide separate and multimodal models for analysing social media content and a comparison of their performance. The study is also expected to identify the benefits and limitations of multimodal feature fusion for this application. The resulting framework and findings can provide a basis for further research in AI-based mental-health monitoring.

### **5.2 Mode of Dissemination**

The outcomes of the project can be documented as a technical research report and presented through an academic project demonstration. If the results provide sufficient research contribution, the work may also be prepared for submission to a suitable conference or journal. Any publication will include proper acknowledgement of datasets, research papers, and other sources used during the project.

### **5.3 Outcome Mapping Table**

The project objectives will be mapped to corresponding technical outcomes such as dataset preparation, unimodal models, multimodal feature fusion, and performance evaluation. The development of the multimodal framework will address the primary objective of combining textual and visual information. Comparative evaluation will provide evidence for assessing the effectiveness of the proposed approach. The final outcomes will therefore directly correspond to the research questions and objectives defined in Chapter 1.

## **5.4 Anticipated Challenges to Dissemination**

Potential challenges include limited availability of suitable labelled multimodal datasets, variations in social media content, and the computational requirements of multimodal models. Differences between textual and visual information may also make feature fusion difficult. Limited project time may restrict extensive experimentation and publication preparation. These challenges will be addressed through careful dataset selection, suitable model selection, and systematic experimentation.

# Chapter 6

## Project Plan and Professional Practices

### 6.1 Timeline and Milestones

**Table 6.1: Project Timeline (Session 2023–2027)**

| Period         | Milestone                                                                         |
|----------------|-----------------------------------------------------------------------------------|
| Jul–Aug 2026   | Literature Review finalized; Research Proposal Synopsis submitted                 |
| Sept 2026      | Detailed system/algorithm design; tool and dataset setup                          |
| Oct 2026       | Implementation / prototyping (Phase I)                                            |
| Nov 2026       | Implementation completion; experimentation and validation                         |
| Dec 2026       | Results analysis; drafting of manuscript for publication                          |
| Jan 2027       | Manuscript/patent draft submission to target venue (e.g., IEEE ICCNT, Sec. 6.2.1) |
| Feb–March 2027 | Peer-review response, revisions, publication and final project report             |

#### 6.1.1 Work Breakdown Structure

The major activities include literature review, dataset preparation, text-model development, image-model development, multimodal integration, testing, performance evaluation, and documentation. These activities will be divided among team members according to their responsibilities. Regular integration of individual components will be performed to ensure that the complete framework functions correctly.

### 6.2 Collaboration and Documentation

The project team will maintain regular communication with the project mentor throughout the development process. Project documents, datasets, code, experimental results, and progress records will be maintained systematically. Version-control and collaborative tools will be used where appropriate to coordinate development and avoid loss of work.

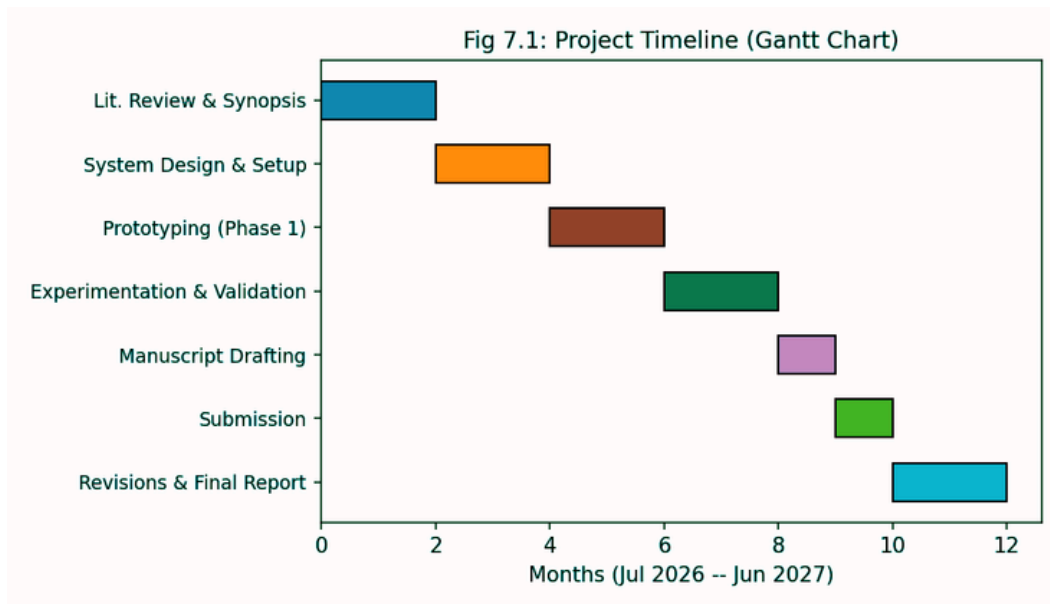


Figure 7.1: Gantt-style project timeline from literature review to final report. (Sample figure – replace with your own dated schedule.)

Table6.2: Team Roles and Responsibilities

| TeamMember | Primary Responsibility                           |
|------------|--------------------------------------------------|
| Kartikey   | Literature and dataset preparation               |
| Akshat     | NLP and text modelling                           |
| Karan      | Image processing, multimodal fusion, evaluation. |
| Lalit      | Documentation                                    |

### 6.2.1 Team Roles and Responsibilities

### 6.2.2 Collaborative Tools and Version Control

GitHub can be used for maintaining and tracking project code and model implementations. Google Drive or similar cloud storage can be used for sharing datasets, documents, and project materials. Canva or other documentation tools may be used for presentation and visual material preparation. Regular backups and version tracking will help maintain consistency throughout the project.